Design and Optimisation of an Automotive Rear Window Defroster

MULTIPHYSICS MODELLING USING COMSOL

ME46095

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Chapter 1

Model Definition

In this project, the design and optimisation of a resistive rear window defroster for an automotive application is investigated using finite element multiphysics modelling in COMSOL Multiphysics 6.2. The defroster must raise the entire glass surface temperature above the dew point to eliminate condensation and frost, while remaining within safe thermal and mechanical stress limits.

1.1 Problem Description

Rear window defrosters operate by passing electric current through thin conductive traces deposited on the inner surface of the glass. Joule heating in the traces conducts into the glass, raising its temperature above the dew point. The key design challenge is to achieve sufficiently uniform heating across the entire window using the minimum number of traces at a voltage compatible with a standard $12\,\mathrm{V}$ automotive electrical system.
The design must satisfy three main criteria:

1. **Thermal:** The minimum surface temperature of the glass must exceed the dew point temperature ($T_{\text{dew}} = 283.15\,\mathrm{K}$, i.e. $10\,^{\circ}\mathrm{C}$).
2. **Structural:** Thermal stresses from differential expansion between the trace material and glass must remain below material yield strengths.
3. **Transient:** The window must defrost within a reasonable time (target: $\leq 15\,\mathrm{min}$).

1.2 Geometry

The model is a two-dimensional representation of a flat rear window. The geometry consists of three components:

- A rectangular glass pane of dimensions $L_{\text{glass}} \times H_{\text{glass}} = 1.5\,\mathrm{m} \times 0.8\,\mathrm{m}$
- Horizontal conductive heating traces spanning the full width of the glass, created using a linear array
- Two vertical bus bars (left and right edges) connecting all traces to the power supply

The number of traces (n_{traces}) is a key design variable, swept from 8 to 22 in the parametric studies. The traces are uniformly spaced across the glass height. Figure 1.1 shows the COMSOL geometry for the baseline configuration.

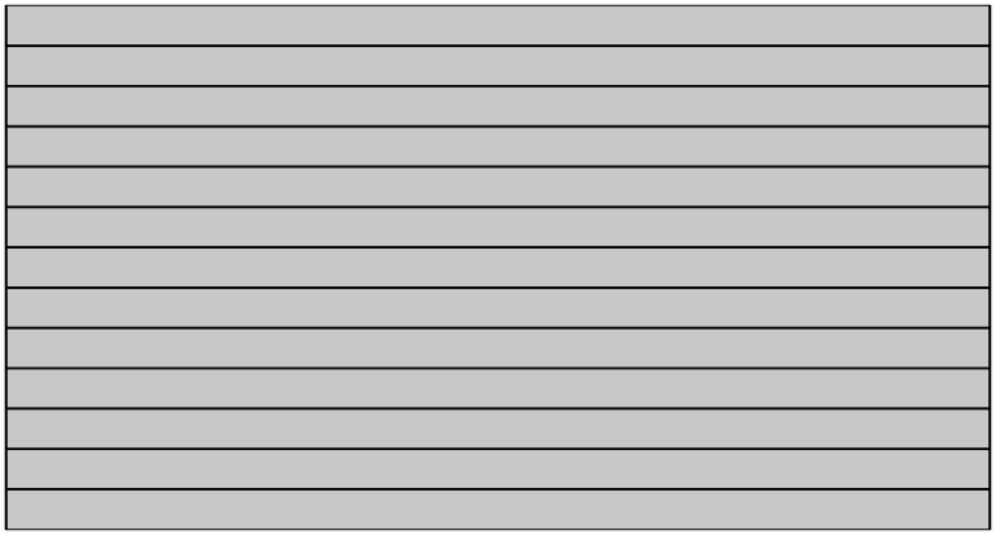


Figure 1.1: Two-dimensional geometry of the rear window defroster model showing the glass pane, horizontal heating traces, and vertical bus bars. Configuration shown: $n_{\text{traces}} = 12$.

The geometric parameters used in COMSOL are listed in Table 1.1.

Table 1.1: Geometric parameters of the rear window defroster model.

Parameter	**Symbol**	**Value**	**Description**
Glass length	L_{glass}	$1.5\,\text{m}$	Width of the window
Glass height	H_{glass}	$0.8\,\text{m}$	Height of the window
Glass thickness	t_{glass}	$5\,\text{mm}$	Out-of-plane thickness
Trace width	w_{trace}	$1.7\,\text{mm}$	Width of each heating trace
Trace thickness	t_{trace}	$0.11\,\text{mm}$	Thickness of each trace
Number of traces	n_{traces}	8–22 (swept)	Design variable

Since the model is two-dimensional with out-of-plane thickness, an effective 2D electrical conductivity is used for the traces:

$$\sigma_{\text{eff}} = \sigma_{\text{trace}} \cdot \frac{t_{\text{trace}}}{t_{\text{glass}}} \tag{1.1}$$

This accounts for the fact that the thin conductive traces occupy only a fraction of the out-of-plane glass thickness.

1.3 Materials

The window glass is modelled as tempered soda-lime glass using COMSOL's built-in material library. The key properties are listed in Table 1.2.

Table 1.2: Material properties of tempered soda-lime glass.

Property	**Value**	**Source**
Thermal conductivity (k)	$1.4\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$	COMSOL library
Density (ρ)	$2500\,\text{kg}\,\text{m}^{-3}$	COMSOL library
Heat capacity (C_p)	$750\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$	COMSOL library
Fracture strength	$120\,\text{MPa}$	Literature

Four candidate trace materials were evaluated, with properties sourced from the CES EduPack database [1]. The full property set is given in Table 1.3.

Table 1.3: Trace material properties from CES EduPack [1]. σ: electrical conductivity, α: coefficient of thermal expansion, E: Young's modulus, ν: Poisson's ratio.

Material	$\boldsymbol{\sigma}$ **(**$\mathrm{S\,m^{-1}}$**)**	$\boldsymbol{\alpha}$ **(**$\mathrm{K^{-1}}$**)**	$\boldsymbol{E}$ **(**GPa**)**	$\boldsymbol{\nu}$	**Yield (**MPa**)**
Pd–Ag alloy	2.3×10^6	15×10^{-6}	102.5	0.38	410
Silver (comm. purity)	5.75×10^7	19.7×10^{-6}	71	0.39	190
Gold (soft annealed)	4.2×10^7	14×10^{-6}	78.5	0.42	20
Nichrome V	9.3×10^5	13×10^{-6}	212.5	0.29	365

1.4 Boundary Conditions

The boundary conditions represent a cold winter morning scenario where the exterior of the glass is exposed to ambient air and the interior faces the vehicle cabin. The values are listed in Table 1.4.

Table 1.4: Boundary conditions and environmental parameters.

Parameter	**Symbol**	**Value**	**Description**
Ambient temperature	T_{amb}	$275.15\,\mathrm{K}$ ($2\,^\circ\mathrm{C}$)	Exterior air
Cabin temperature	T_{cabin}	$293.15\,\mathrm{K}$ ($20\,^\circ\mathrm{C}$)	Interior cabin air
Initial temperature	T_{init}	$278.15\,\mathrm{K}$ ($5\,^\circ\mathrm{C}$)	Frosted window (transient)
Dew point temperature	T_{dew}	$283.15\,\mathrm{K}$ ($10\,^\circ\mathrm{C}$)	Target minimum temperature
External heat transfer coeff.	h_{ext}	$15\,\mathrm{W\,m^{-2}\,K^{-1}}$	Exterior convection
Internal heat transfer coeff.	h_{int}	$7\,\mathrm{W\,m^{-2}\,K^{-1}}$	Interior convection
Applied voltage	V_{in}	10–$14.5\,\mathrm{V}$ (swept)	Automotive 12V system range

Electrical. An electric potential V_{in} is applied on the left bus bar, and ground ($V = 0$) on the right bus bar. The electric current physics is defined only on the trace and bus bar domains (not the glass, which is electrically insulating).
Thermal. Since the model is 2D with out-of-plane thickness, heat losses to the exterior and interior are modelled using a volumetric heat source that accounts for convection on both sides of the glass:

$$Q_{\text{conv}} = \frac{h_{\text{ext}}(T_{\text{amb}} - T) + h_{\text{int}}(T_{\text{cabin}} - T)}{t_{\text{glass}}} \tag{1.2}$$

This expression replaces the out-of-plane heat flux boundary condition, which caused spatial frame conflicts in COMSOL 6.2 when coupled with solid mechanics.
Mechanical. For the stress analysis (Sim 2c), fixed constraints are applied on the left and right outer edges of the glass, representing the car body frame holding the window. This is a conservative simplification; in reality, a rubber seal provides some compliance.

1.5 Physics Coupling and Governing Equations

The model involves three coupled physics: Electric Currents, Heat Transfer in Solids, and Solid Mechanics. Their coupling is illustrated in Figure 1.2.

TODO: Create flow diagram — EC → Joule Heating → HT → Thermal Expansion → SM (draw manually)

Figure 1.2: Physics coupling in the multiphysics model. Joule heating from Electric Currents drives Heat Transfer, which in turn drives thermal expansion in Solid Mechanics.

Electric Currents. The electric field in the conductive traces is governed by current conservation:

$$\nabla \cdot \mathbf{J} = 0 \tag{1.3}$$

where $\mathbf{J} = \sigma_{\text{eff}}\mathbf{E}$ is the current density and $\mathbf{E} = -\nabla V$ is the electric field.
Heat Transfer in Solids. The heat equation in the glass domain is:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + Q_e + Q_{\text{conv}} \tag{1.4}$$

where $Q_e = \mathbf{J} \cdot \mathbf{E}$ is the resistive (Joule) heating from the Electromagnetic Heating multiphysics coupling, and Q_{conv} is the convective heat source from Equation (1.2).
For stationary studies, the left-hand side is zero and the equation reduces to:

$$\nabla \cdot (k\nabla T) + Q_e + Q_{\text{conv}} = 0 \tag{1.5}$$

Solid Mechanics. The mechanical equilibrium is described by:

$$\nabla \cdot \mathbf{S} + \mathbf{F}_v = 0 \tag{1.6}$$

where $\mathbf{S}$ is the Cauchy stress tensor. Thermal expansion introduces an initial strain:

$$\varepsilon_{\text{th}} = \alpha(T - T_{\text{ref}})\mathbf{I} \tag{1.7}$$

with reference temperature $T_{\text{ref}} = T_{\text{amb}} = 275.15\,\mathrm{K}$.

Chapter 2

Model Implementation in COMSOL

The design was developed through five simulations each building on the previous one, progressively refining the design from a baseline model to a optimised solution. Table 2.1 summarises all simulations.

Table 2.1: Overview of simulations performed.

Simulation	Study Type	Sweep Parameters	Objective
Sim 1	Stationary	None (baseline)	Verify model physics
Sim 2	Stationary	V_{in}, n_{traces}	Identify viable operating range
Sim 2b	Stationary	Material, n_{traces}, V_{in}	Material optimisation
Sim 2c	Stationary (sequential)	Material, n_{traces}	Stress analysis
Sim 3	Time dependent	Material, n_{traces}, V_{in}	Defrost time
Sim 4	Time dependent	Zone layout (n_{top}, n_{mid}, n_{bot})	Zoned trace spacing
Sim 5	Time dependent	Chebyshev exponent p	Non-uniform spacing optimisation

2.1 Simulation 1 — Baseline Stationary Model

The first simulation uses the default model setup: 12 silver-paste traces ($\sigma = 2 \times 10^6\,\mathrm{S\,m^{-1}}$) with an input voltage of $V_{\text{in}} = 12\,\mathrm{V}$. A stationary study is used, combining Electric Currents and Heat Transfer through Electromagnetic Heating.
Sim 1 is mainly used as a first check that the model behaves as expected and the firstly assumed amount of traces are somewhat realistic. The voltage should drop from $12V$ at the left bus bar to $0V$ at the right bus bar. The temperature should stay within a realistic range of about 270 to $320K$, and the coldest regions should appear near the top and bottom edges, where there are no traces to enclose the space. Traces were not placed at the top and bottom edges of the glass, as placing traces there would trivially reduce cold spots since more traces equal more heat. The purpose of this study is to optimize the trace distribution pattern within a fixed heated region, showing that the performance can be increased significantly through pattern optimization.

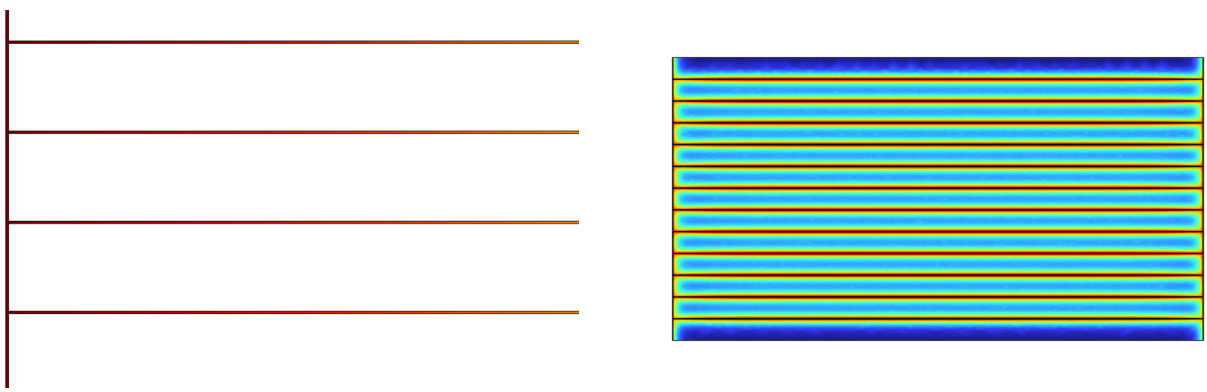


Figure 2.1: Sim 1 baseline results at $V_{\text{in}} = 12\,\mathrm{V}$, $n_{\text{traces}} = 12$, silver paste. (a) Electric potential distribution on the traces and bus bars. (b) Temperature distribution across the glass surface.

The baseline results show a surface-average temperature of $297.13\,\mathrm{K}$ (above the dew point) but a surface-

minimum temperature of $282.12\,\mathrm{K}$ (below the dew point). The cold spots are at the top and bottom edges of the glass where trace coverage does not extend. These values are realistic and now it is possible to start looking at further optimization.

2.2 Simulation 2 — Voltage and Trace Count Sweeps

Sim 2b studies how temperature changes with input voltage and the number of traces in order to first find the optimal number of traces. Important to state is that these first simulations are stationary studies, so they do not give any information about how long it takes to defrost the window, but they do give information about the minimum temperature that can be reached for each configuration.
Sim 2 then examines how the minimum glass temperature depends on voltage and trace count, both separately and in combination.
A parametric sweep of V_{in} from $8\,\mathrm{V}$ to $16\,\mathrm{V}$ (9 values) is performed at $n_{\text{traces}} = 12$. Even at $16\,\mathrm{V}$, the minimum temperature ($283.09\,\mathrm{K}$) barely reaches the dew point, indicating that voltage alone cannot solve the problem with 12 traces.
A parametric sweep of n_{traces} from 8 to 22 is performed at $V_{\text{in}} = 12\,\mathrm{V}$. Results are shown in Figure 2.2.

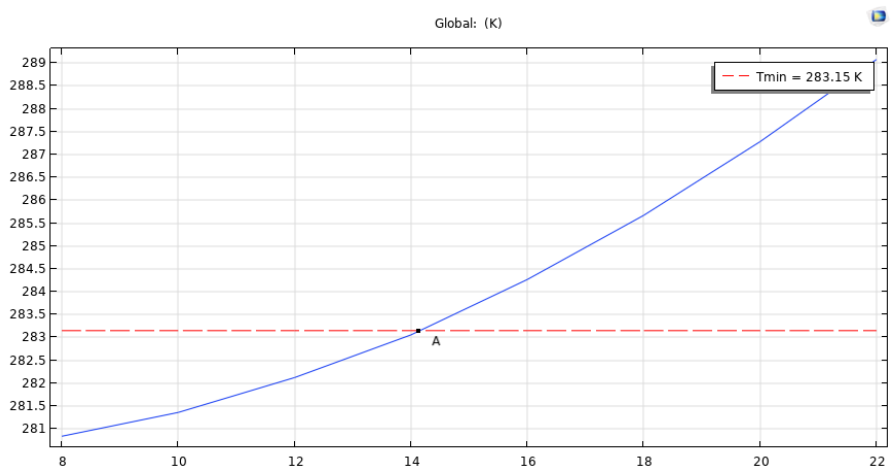


Figure 2.2: Minimum glass surface temperature as a function of the number of traces at $V_{\text{in}} = 12\,\mathrm{V}$. The horizontal dashed line indicates the dew point temperature ($283.15\,\mathrm{K}$) that crosses the curve at point A indicating the amount of traces needed to heat up the window to dew temperature.

Figure 2.2 shows that the minimum glass temperature crosses the dew point at slightly more than 14 traces. To ensure a safe margin well above the dew temperature, $n_{\text{traces}} = 16$ is chosen as the baseline for the next simulations.
A full factorial sweep over $n_{\text{traces}} \in \{8, 10, 12, 14, 16, 18, 20, 22\}$ and $V_{\text{in}} \in \{8, 10, 12, 14, 16\}$ V produces 35 stationary solutions. Results are shown in Table 2.2.

Table 2.2: Minimum surface temperature T_{min} (K) for the combined voltage–trace count sweep (silver paste, $\sigma = 2 \times 10^6\,\mathrm{S\,m^{-1}}$). Values $\geq 283.15\,\mathrm{K}$ are shown in **bold**.

n_{traces}	$8\,\mathrm{V}$	$10\,\mathrm{V}$	$12\,\mathrm{V}$	$14\,\mathrm{V}$	$16\,\mathrm{V}$
8	280.86	280.85	280.84	280.83	280.81
10	281.09	281.21	281.36	281.54	281.74
12	281.43	281.74	282.12	282.57	283.09
14	281.85	282.39	283.06	**283.84**	**284.75**
16	282.38	**283.23**	**284.27**	**285.49**	**286.91**
18	283.01	**284.21**	**285.67**	**287.40**	**289.40**
20	**283.72**	**285.33**	**287.28**	**289.60**	**292.27**
22	**284.52**	**286.57**	**289.08**	**292.04**	**295.46**

The minimum configuration in this sweep that remains above the dew point is 16 traces at 10 V, yielding a minimum surface temperature of 283.23 K. The results also show that the effect of increasing voltage becomes much stronger as the number of traces increases, indicating that the effects of voltage and trace count are coupled.

2.3 Simulation 2b — Material Optimisation

Sim 2b extends the parametric sweep to include four candidate trace materials from the CES EduPack database [1]. The sweep covers $4 \times 8 \times 4 = 128$ combinations (4 materials, 8 trace counts, 4 voltages).
The voltages used reflect the realistic automotive range: $10\,\mathrm{V}$ (battery low), $12\,\mathrm{V}$ (nominal), $13.5\,\mathrm{V}$ (engine running), and $14.5\,\mathrm{V}$ (maximum alternator output).
The minimum number of traces needed to reach the dew point for each material and voltage is shown in Table 2.3.

Table 2.3: Minimum number of traces to reach $T_{\text{min}} \geq 283.15\,\mathrm{K}$ for each material and voltage.

Material	$10\,\mathrm{V}$	$12\,\mathrm{V}$	$13.5\,\mathrm{V}$	$14.5\,\mathrm{V}$
Pd–Ag alloy	16	14	14	14
Nichrome V	22	18	18	16
Silver (comm. purity)	*Overheats — not viable*			
Gold (soft annealed)	*Overheats — not viable*			

The full temperature results for the two viable materials (Pd–Ag alloy and Nichrome V) are shown in Tables 2.4 and 2.5.

Table 2.4: T_{min} (K) for Pd–Ag alloy ($\sigma = 2.3 \times 10^6\,\mathrm{S\,m^{-1}}$). Values $\geq 283.15\,\mathrm{K}$ shown in **bold**.

n_{traces}	$10\,\mathrm{V}$	$12\,\mathrm{V}$	$13.5\,\mathrm{V}$	$14.5\,\mathrm{V}$
8	280.85	280.83	280.82	280.82
10	281.26	281.43	281.58	281.69
12	281.87	282.31	282.69	282.97
14	282.62	**283.38**	**284.05**	**284.54**
16	**283.59**	**284.78**	**285.81**	**286.57**
18	**284.71**	**286.39**	**287.86**	**288.93**
20	**286.00**	**288.24**	**290.20**	**291.63**
22	**287.43**	**290.31**	**292.82**	**294.65**

Table 2.5: T_{min} (K) for Nichrome V ($\sigma = 9.3 \times 10^5\,\mathrm{S\,m^{-1}}$). Values $\geq 283.15\,\mathrm{K}$ shown in **bold**.

n_{traces}	$10\,\mathrm{V}$	$12\,\mathrm{V}$	$13.5\,\mathrm{V}$	$14.5\,\mathrm{V}$
8	280.87	280.86	280.86	280.85
10	281.03	281.10	281.16	281.21
12	281.28	281.46	281.61	281.72
14	281.58	281.89	282.16	282.36
16	281.97	282.45	282.87	**283.18**
18	282.43	**283.11**	**283.70**	**284.13**
20	282.95	**283.86**	**284.65**	**285.23**
22	**283.53**	**284.69**	**285.71**	**286.45**

TODO (Sebtain): Plot — T_{min} vs n_{traces} for all 4 materials at $V_{in} = 12\,\mathrm{V}$ with dew point line

Figure 2.3: Comparison of minimum surface temperature vs. number of traces for all four EduPack materials at $V_{\text{in}} = 12\,\mathrm{V}$. Silver and Gold exceed safe temperatures at $n_{\text{traces}} \geq 10$ and are not viable.

- **Silver (commercial purity) and Gold (soft annealed) are not viable** — their bulk conductivity ($> 10^7$ S/m) is too high for the screen-printed geometry, causing extreme overheating (Silver reaches $625\,\mathrm{K}$ at 22 traces, $14.5\,\mathrm{V}$).
- **Pd–Ag alloy is the best EduPack material** — it reaches the dew point at 14 traces ($12\,\mathrm{V}$+). Temperatures remain in a safe range.
- **Nichrome V is viable but requires more traces** (18–22 depending on voltage).

2.4 Simulation 2c — Stress Analysis

Sim 2c evaluates the thermal stresses for the two shortlisted materials (Pd–Ag alloy and Nichrome V). Solid Mechanics and Thermal Expansion are re-enabled, and a sequential solve is used:

- **Step 1:** Electric Currents + Heat Transfer (stationary)
- **Step 2:** Solid Mechanics with Thermal Expansion (stationary, using Step 1 temperatures)

This sequential approach avoids the spatial frame conflict ("dvol_spatial undefined") encountered when solving all physics simultaneously in COMSOL 6.2.

Table 2.6: Von Mises stress results at $V_{\text{in}} = 12\,\mathrm{V}$.

Material	n_{traces}	**Max stress (MPa)**	**Yield (MPa)**	**Safety factor**
Pd–Ag alloy	14	120.3	410	3.4×
Pd–Ag alloy	18	134.5	410	3.0×
Nichrome V	14	94.6	365	3.9×
Nichrome V	18	102.9	365	3.5×

TODO (Sebtain): Sim2c — von Mises stress, Pd–Ag, 14 traces

TODO (Sebtain): Sim2c — von Mises stress, Nichrome V, 18 traces

Figure 2.4: Von Mises stress distribution at $V_{\text{in}} = 12\,\mathrm{V}$. (a) Pd–Ag alloy, $n_{\text{traces}} = 14$. (b) Nichrome V, $n_{\text{traces}} = 18$. Peak stresses concentrate at trace endpoints near the fixed edges.

Both materials pass the stress constraint with safety factors $> 3\times$. Peak stresses ($\sim 120\,\mathrm{MPa}$) concentrate at trace endpoints near the fixed boundary — a conservative artifact of the fully fixed edge condition. The glass bulk stress remains very low ($2\,\mathrm{MPa}$ to $4\,\mathrm{MPa}$), well below the $120\,\mathrm{MPa}$ fracture strength of tempered glass.

2.5 Simulation 3 — Transient Defrost Study

Sim 3 evaluates how long each configuration takes to defrost the window from the initial frosted condition ($T_{\text{init}} = 278.15\,\mathrm{K}$) to the point where $T_{\text{min}} \geq T_{\text{dew}}$.
A time-dependent study is used with output times from 0 to $3600\,\mathrm{s}$ in $60\,\mathrm{s}$ increments. The Electric Currents equation is set to stationary form (the electric field equilibrates in microseconds compared to the thermal time scale of minutes). Solid Mechanics is disabled for computational efficiency.
The sweep covers: 2 materials $\times$ 5 trace counts (14, 16, 18, 20, 22) $\times$ 2 voltages (12, $14.5\,\mathrm{V}$) = 20 runs. The mesh was refined to Fine (189,192 domain elements) for accuracy.

TODO (Sebtain): Plot — T_{min} vs time for Pd–Ag at $12\,\mathrm{V}$, curves for $n = 14, 16, 18, 20, 22$ with dew point line

Figure 2.5: Transient evolution of minimum surface temperature for Pd–Ag alloy at $V_{\text{in}} = 12\,\mathrm{V}$ for different numbers of traces. The horizontal dashed line indicates the dew point.

2.6 Simulation 4 — Trace Pattern Optimisation

Now that the optimal material (Pd–Ag alloy) has been determined with its operating conditions, it is possible to look at the optimal trace pattern which is the scope of this project. The optimal trace pattern is one where after heating heats up the glass as uniformly as possible in the shortest time possible. Although an obvious way to improve heating performance would be to increase both the number of traces and the applied voltage,

this lies outside the scope of the project, since the aim is to optimize the pattern itself instead of overall input parameters

Sim 4 investigates the effect of trace layout on the temperature uniformity across the glass surface. The baseline simulations revealed that cold spots form at the top and bottom edges of the glass, farthest from any heating trace. Therefore a logical first strategy would be to increase the density of traces neaar the edges to improve the heating.

This has been done for the first simulation and the glass was divided into three horizontal zones with independent trace densities Figure 2.6. Zone parameters were added to Global Definitions and each zone was created as a separate Rectangle + Array in the geometry, with spacing $H_{\text{zone}}/(n_{\text{zone}}+1)$. :

- **Top zone:** $H_{\text{top}} = 200\,\text{mm}$, $n_{\text{top}} = 6$ traces
- **Middle zone:** $H_{\text{mid}} = 400\,\text{mm}$, $n_{\text{mid}} = 8$ traces
- **Bottom zone:** $H_{\text{bot}} = 200\,\text{mm}$, $n_{\text{bot}} = 6$ traces

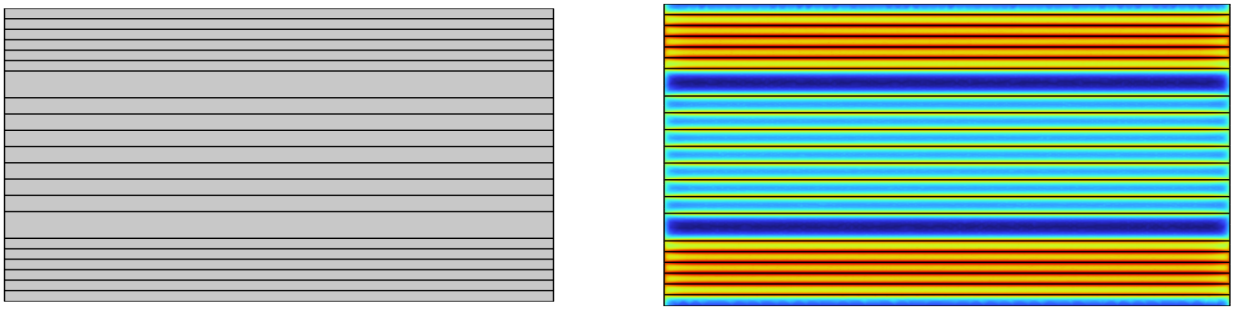


Figure 2.6: Zoned trace layout (Approach A). (a) Geometry with 6/8/6 distribution across top/middle/bottom zones. (b) Steady-state temperature distribution showing reduced cold spots at the edges.

The study was changed from stationary to time dependent with output range `range(0,60,3600)`, i.e. every $60\,\text{s}$ for $60\,\text{min}$, the 6/8/6 zoned configuration reached T_{dew} $(283.15\,\text{K})$ at $t = 660\,\text{s} = 11\,\text{min}$. The Electric Currents equation was solved in stationary form (as in Sim 3), because the electric field reaches equilibrium much faster than the temperature changes. The initial temperature was set to $T_{\text{init}} = 278.15\,\text{K}$. The zoned approach was not analysed further because the continuous Chebyshev spacing (Sim 5) turned out to be more effective, as discussed in Section 2.7.

2.7 Simulation 5 — Chebyshev Trace Spacing Optimisation

While the zoned approach (Sim 4) shows improvement in the heating for the outer edges it is still not optimal and it is based on personal intuition of the distribution for the traces with no real mathematical backing. A mathematical alternative for the trace spacing is to use a continuous spacing function that clusters traces near the edges. Chebyshev node spacing, derived from the roots of Chebyshev polynomials, is known in numerical analysis for minimising interpolation error by placing points near interval boundaries. This property can be used for the window is design goal of putting heating where it is most needed.

2.7.1 Spacing Formula and Implementation

The vertical position y_i of the i-th trace ($i = 1, \ldots, 20$) is given by:

$$y_i = \frac{H_{\text{glass}}}{2}\left(1 - \text{sign}\left(\cos\frac{i\pi}{21}\right)\left|\cos\frac{i\pi}{21}\right|^p\right) \tag{2.1}$$

The inner cosine term $\cos\left(\frac{i\pi}{n_{\text{traces}}+1}\right)$ produces Chebyshev-type nodes between -1 and $+1$ that are inherently denser near the extremes [2]. The outer factor $\frac{H_{\text{glass}}}{2}(1-\cdot)$ maps these values onto the full glass height. The exponent p shapes the distribution: for $p > 1$, the trace clustering increases at the centre of the glass and decreases at the edges, for $p < 1$, the centre gap widens and the traces at the edge increase and eventually leading to uniform spacing as $p \to 0$.

Because raising a negative number to a non-integer power is impossible, the $\text{sign}(\cdot)|\cdot|^p$ formulation separates sign and magnitude, preserving symmetry for all real p.

In COMSOL, the traces were positioned at the y_i computed from Equation (2.1). The exponent p was defined as a global parameter so that it could be used in a parametric sweep and it became possible to automatically regenerate the geometry and re-solve for each value of p. All other model settings (material: Pd–Ag alloy, σ_{eff} correction, time-dependent study, boundary conditions) were kept identical to Sim 4.

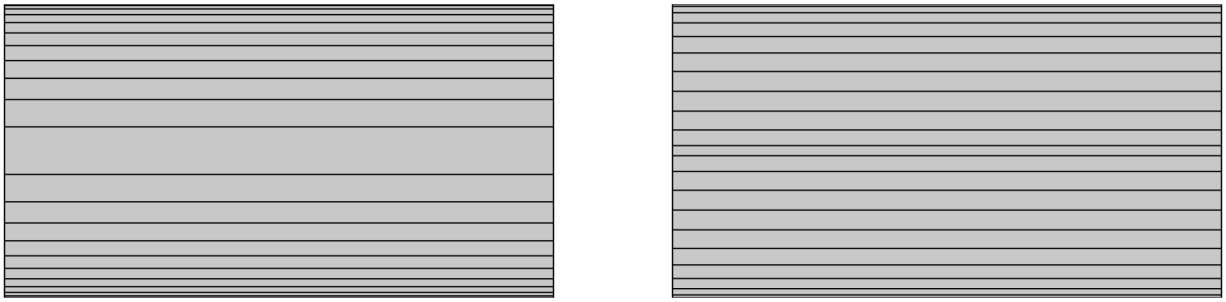


Figure 2.7: Chebyshev trace spacing for (a) $p = 0.7$ (stronger edge clustering, wider centre gap) and (b) $p = 1.3$ (optimal). Centre gaps decrease with increasing p while edge gaps widen.

2.7.2 Results and Optimum

Three values of p were tested first to establish the feasibility of the approach:

- $p = 1.0$ (standard Chebyshev): **defrost in** $7\,\text{min}$, steady-state $T_{\text{min}} = 292.6\,\text{K}$
- $p = 0.85$: **never reaches dew point** — steady-state $T_{\text{min}} = 282.06\,\text{K}$
- $p = 0.7$: **never reaches dew point** — steady-state $T_{\text{min}} = 282.06\,\text{K}$

Already for $p = 1.0$ an improvement can be noticed with a time reduction from $12\,\text{min}$ (uniform) to $7\,\text{min}$, a 42% improvement with the same number of traces, same material, and same voltage. However, reducing p below 1.0 removes too much edge clustering and the cold spots return.
To find the optimal exponent, a parametric sweep was performed for various values for p, the results are presented in Table 2.7 and plotted in Figure 2.8.

Table 2.7: Chebyshev exponent sweep results (Pd–Ag alloy, 20 traces, $V_{\text{in}} = 12\,\text{V}$). $T_{360\text{s}}$: minimum temperature at $t = 6\,\text{min}$.

p	T_{min} at $t = 360\,\text{s}$ (K)	**Defrost time**	**Steady-state** T_{min} (K)
0.70	279.53	Never	282.1
0.80	279.72	23 min	284.0
0.90	280.44	12 min	287.5
1.00	282.23	8 min	292.6
1.10	282.97	7 min	294.5
1.20	283.16	6 min	294.9
1.30	**283.20**	**6 min**	**295.0**
1.40	283.09	6 min	294.8
1.50	282.97	7 min	294.5
1.60	282.91	7 min	294.4
1.70	282.74	7 min	293.9
1.80	282.56	7 min	293.4
1.90	282.40	7 min	293.1
2.00	282.28	7 min	292.8
3.00	281.17	9 min	289.9

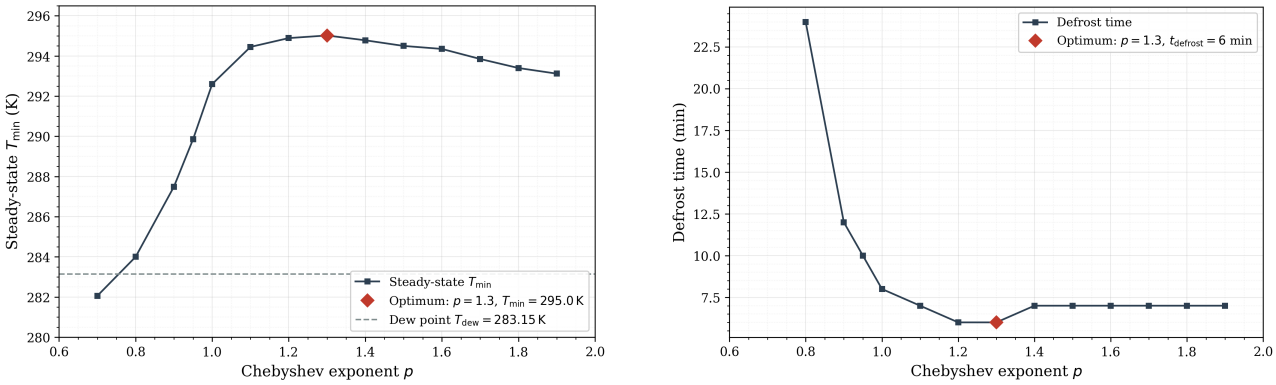


Figure 2.8: Effect of the Chebyshev exponent p on (a) steady-state minimum temperature and (b) defrost time. The optimum is $p = 1.3$, achieving the highest T_{min} and fastest defrost.

The optimum is clearly at $p = 1.3$, see Figure 2.8: it achieves the highest minimum temperature at the 6-minute mark ($283.20\,\text{K}$) and the highest steady-state T_{min} ($295.0\,\text{K}$) as can be seen in Figure 2.9. Since the Chebyshev function is a periodic function it is not said that increasing or decreasing the p-value will lead to optimal behaviour. The same pattern will repeat itself for higher values of p but the wavelength will increase and the clusters are not visible anymore for the respective window height.

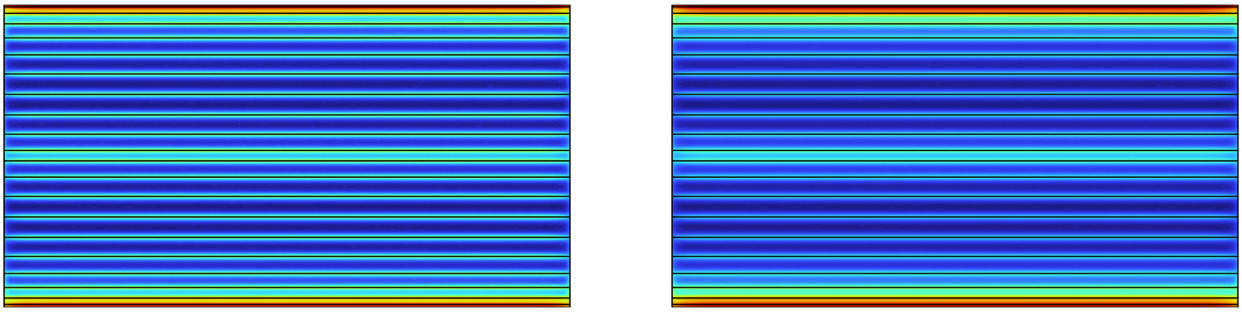


Figure 2.9: Temperature distribution for Chebyshev spacing $p = 1.3$ (Pd–Ag alloy, 20 traces, $12\,\text{V}$). (a) At $t = 6\,\text{min}$ (defrost moment): $T_{\text{min}} = 283.20\,\text{K}$. (b) Steady state: $T_{\text{min}} = 295.0\,\text{K}$.

2.7.3 Practical Considerations

An important practical constraint for rear window trace patterns is driver visibility. Per the SAE J216 standard, obscuration of the rear vision area should not exceed 5% in the primary zone [3]. For 20 traces with $w_{\text{trace}} = 1.7\,\text{mm}$:

$$\text{Obscuration} = \frac{n_{\text{traces}} \times w_{\text{trace}}}{H_{\text{glass}}} = \frac{20 \times 1.7}{800} = 4.25\% \tag{2.2}$$

This is well within the 5% limit regardless of the spacing pattern, since the total trace area is unchanged. The Chebyshev spacing only redistributes the same number of traces, so visibility is not affected. Additionally, UNECE Regulation No. 43 requires a minimum light transmittance of 70% for safety glazing in vehicles [4], which is easily satisfied given the low obscuration ratio.

Check if this is necessary to add because it indicates something we didnt do out of laziness

As a further form of optimization it was considered to add vertical traced connecting the horizontal ones. Adding vertical traces between the horizontal ones was considered to create a grid pattern. However, analysis showed this would be ineffective: since all parallel horizontal traces are at the same voltage at any given x-position, vertical connectors between them would have zero potential difference ($\Delta V = 0$) and carry no current. Variable-width traces could create voltage differences enabling vertical current flow, but this adds significant manufacturing complexity without clear benefit over optimising the horizontal spacing alone.

Chapter 3

Model Validation

3.1 Mesh Convergence Study

A mesh convergence study was performed to ensure that the results are independent of the mesh resolution. The transient defrost simulation (Sim 3) was run with two mesh settings: Extra coarse and Fine. Table 3.1 compares the results for representative configurations.

Table 3.1: Mesh convergence results for the transient defrost study (Pd–Ag alloy, $V_{\text{in}} = 12\,\text{V}$).

		Extra coarse mesh	**Fine mesh** (189,192 elements)
$n_{\text{traces}} = 14$	Defrost time	42 min	41 min
	T_{min} (steady)	283.38 K	283.38 K
$n_{\text{traces}} = 20$	Defrost time	12 min	12 min
	T_{min} (steady)	288.24 K	288.24 K

TODO (Sebtain): Extra coarse mesh

TODO (Sebtain): Fine mesh (189,192 elements)

Figure 3.1: Comparison of mesh densities used in the convergence study. (a) Extra coarse mesh. (b) Fine mesh (189,192 domain elements).

For the Pd–Ag alloy configurations, the stationary temperature results are virtually identical between the two mesh levels, and the transient defrost times agree within 1 minute. This confirms that the Fine mesh provides converged results.
However, the mesh refinement did reveal an important difference for the Nichrome V material near the dew point boundary:

- **Nichrome V, 18 traces, $12\,\text{V}$:** The coarse mesh predicted a defrost time of 48 minutes, while the fine mesh shows that the steady-state T_{min} is only $283.09\,\text{K}$ — *below* the dew point. The window never fully defrosts.
- **Nichrome V, 16 traces, $14.5\,\text{V}$:** The coarse mesh predicted "Never", while the fine mesh gives 59 minutes ($T_{\text{min}} = 283.152\,\text{K}$, barely above the dew point).

These borderline cases highlight the importance of sufficient mesh resolution when results are near a threshold.

3.2 Comparison to Literature

The model physics and boundary condition values are validated against published experimental and numerical studies.
The convection coefficients used in this model ($h_{\text{ext}} = 15\,\mathrm{W\,m^{-2}\,K^{-1}}$, $h_{\text{int}} = 7\,\mathrm{W\,m^{-2}\,K^{-1}}$) are compared with experimental measurements by Hożejowska et al. [5], who reported measured h values for automotive glass under various driving conditions. Their results support the order of magnitude used in this study, with exterior values in the range $10\,\mathrm{W\,m^{-2}\,K^{-1}}$ to $25\,\mathrm{W\,m^{-2}\,K^{-1}}$ depending on vehicle speed.
Rühl et al. [6] performed coupled finite element simulations of electrically heated glass panes with resistive lines, validated against infrared thermography and thermocouple measurements. Their modelling approach — coupling electric currents with heat transfer via Joule heating in a 2D representation — is directly analogous to the physics setup used in this work.
Koshlak et al. [7] measured that 83–85% of the heat generated in electrically heated automotive glass is directed inward (toward the cabin side), which is consistent with the asymmetric convection coefficients used in this model ($h_{\text{int}} < h_{\text{ext}}$, meaning less heat loss to the interior).

3.3 Back-of-Envelope Power and Temperature Estimate

An independent analytical estimate of the steady-state average temperature provides a direct check on the numerical results. The approach is to compute the total Joule heating power from the trace resistance and compare the resulting energy balance temperature with the COMSOL output.
Each trace has electrical resistance:

$$R_{\text{trace}} = \frac{L_{\text{glass}}}{\sigma_{\text{trace}} \cdot w_{\text{trace}} \cdot t_{\text{trace}}} \tag{3.1}$$

For the Sim 1 baseline (silver paste, $\sigma = 2 \times 10^6\,\mathrm{S\,m^{-1}}$):

$$R_{\text{trace}} = \frac{1.5}{2 \times 10^6 \times 1.7 \times 10^{-3} \times 0.11 \times 10^{-3}} = 4.01\,\Omega \tag{3.2}$$

The power per trace is $P_{\text{trace}} = V_{\text{in}}^2/R_{\text{trace}} = 144/4.01 = 35.9\,\mathrm{W}$, and the total power for 12 traces is:

$$P_{\text{total}} = 12 \times 35.9 = 431\,\mathrm{W} \tag{3.3}$$

Note that this physical power is correctly reproduced in the 2D COMSOL model through the σ_{eff} correction (Equation (1.1)): the effective resistance in the model is $R_{\text{model}} = L_{\text{glass}}/(\sigma_{\text{eff}} \cdot w_{\text{trace}} \cdot t_{\text{glass}}) = L_{\text{glass}}/(\sigma_{\text{trace}} \cdot w_{\text{trace}} \cdot t_{\text{trace}}) = R_{\text{trace}}$.
At steady state, the total Joule heating must equal the convective losses from both sides of the glass:

$$P_{\text{total}} = (h_{\text{ext}} + h_{\text{int}}) \cdot A_{\text{glass}} \cdot (T_{\text{avg}} - T_{\text{weighted}}) \tag{3.4}$$

where $A_{\text{glass}} = L_{\text{glass}} \times H_{\text{glass}} = 1.2\,\mathrm{m}^2$ and the convection-weighted ambient temperature is:

$$T_{\text{weighted}} = \frac{h_{\text{ext}} \cdot T_{\text{amb}} + h_{\text{int}} \cdot T_{\text{cabin}}}{h_{\text{ext}} + h_{\text{int}}} = \frac{15 \times 275.15 + 7 \times 293.15}{22} = 280.9\,\mathrm{K} \tag{3.5}$$

Solving for T_{avg}:

$$T_{\text{avg}} = T_{\text{weighted}} + \frac{P_{\text{total}}}{(h_{\text{ext}} + h_{\text{int}}) \cdot A_{\text{glass}}} = 280.9 + \frac{431}{22 \times 1.2} = 280.9 + 16.3 = 297.2\,\mathrm{K} \tag{3.6}$$

The COMSOL surface-average temperature for Sim 1 (12 traces, silver paste, $12\,\mathrm{V}$) is $T_{\text{avg}} = 297.13\,\mathrm{K}$. The back-of-envelope estimate gives $297.2\,\mathrm{K}$ — a deviation of only $0.07\,\mathrm{K}$ ($< 0.03\%$). This excellent agreement confirms that:

- The σ_{eff} correction correctly maps the 3D trace geometry to the 2D model
- The Joule heating and convective boundary conditions are consistently implemented
- The total energy balance of the model is accurate

Table 3.2: Back-of-envelope validation of the average temperature for selected configurations.

Material	n_{traces}	P_{total} (W)	$T_{\text{avg}}^{\text{analytical}}$ (K)	$T_{\text{avg}}^{\text{COMSOL}}$ (K)	**Error**
Silver paste	12	431	297.2	297.13	0.03%
Pd–Ag alloy	14	578	302.8	— *extract from COMSOL*	—

STILL DO: EXTRACT T_{avg} FROM COMSOL FOR Pd–Ag 14-TRACE AND 20-TRACE CONFIGURATIONS TO COMPLETE THE TABLE

3.4 Sensitivity Analysis

The parametric sweeps performed in Sim 2 and Sim 2b serve as a de facto sensitivity analysis, showing how the minimum surface temperature responds to changes in the three main design variables.

Table 3.3: Sensitivity of minimum surface temperature to design parameters (Pd–Ag alloy, baseline: 14 traces, $12\,\mathrm{V}$).

Parameter change	ΔT_{min} **(K)**	**Sensitivity**
n_{traces}: 14 $\rightarrow$ 16 (+2 traces)	+1.40	$0.70\,\mathrm{K}$/trace
V_{in}: 12 $\rightarrow$ $14.5\,\mathrm{V}$ (+$2.5\,\mathrm{V}$)	+1.16	$0.46\,\mathrm{K\,V^{-1}}$
Material: Pd–Ag $\rightarrow$ Nichrome V	-1.49	—

The results confirm that **the number of traces is the most influential parameter**, followed by applied voltage. The choice of trace material has a strong effect through its conductivity: higher conductivity produces more Joule heating per unit length of trace.
The external heat transfer coefficient h_{ext} is also a sensitive parameter (as noted in the literature [5]), since it directly controls the rate of heat loss from the glass. However, this was not systematically swept in the current study and is recommended as future work.

Chapter 4

Results and Discussion

4.1 Transient Defrost Performance

SEBTAIN PART
The transient defrost times for all tested configurations are presented in Tables 4.1 and 4.2. These results are from the fine-mesh simulation (189,192 domain elements).

Table 4.1: Defrost time (minutes) for Pd–Ag alloy ($\sigma = 2.3 \times 10^6\,\mathrm{S\,m^{-1}}$). Defrost is defined as the time for T_{min} to reach $283.15\,\mathrm{K}$.

n_{traces}	$12\,\mathrm{V}$	$14.5\,\mathrm{V}$
14	41	22
16	22	16
18	16	12
20	**12**	10
22	10	**8**

Table 4.2: Defrost time (minutes) for Nichrome V ($\sigma = 9.3 \times 10^5\,\mathrm{S\,m^{-1}}$). “Never” indicates that T_{min} does not reach $283.15\,\mathrm{K}$ within $60\,\mathrm{min}$.

n_{traces}	$12\,\mathrm{V}$	$14.5\,\mathrm{V}$
14	Never	Never
16	Never	59
18	Never	25
20	28	18
22	20	14

TODO (Sebtain): Bar chart — defrost time comparison, Pd–Ag vs Nichrome V at $12\,\mathrm{V}$ *and* $14.5\,\mathrm{V}$

Figure 4.1: Defrost time comparison between Pd–Ag alloy and Nichrome V at $12\,\mathrm{V}$ and $14.5\,\mathrm{V}$. "N" indicates the configuration never reaches the dew point within $60\,\mathrm{min}$.

Pd–Ag alloy consistently defrosts faster than Nichrome V at the same trace count and voltage. Nichrome V requires ≥ 20 traces at $12\,\mathrm{V}$ to defrost at all within 60 minutes, while Pd–Ag defrosts with as few as 14 traces (in 41 minutes).

4.2 Temperature Distribution

CHECK AGAIN WRON PIC
Figure 4.2 shows the steady-state temperature distribution for the recommended design (Pd–Ag alloy, 20 traces, $12\,\mathrm{V}$).

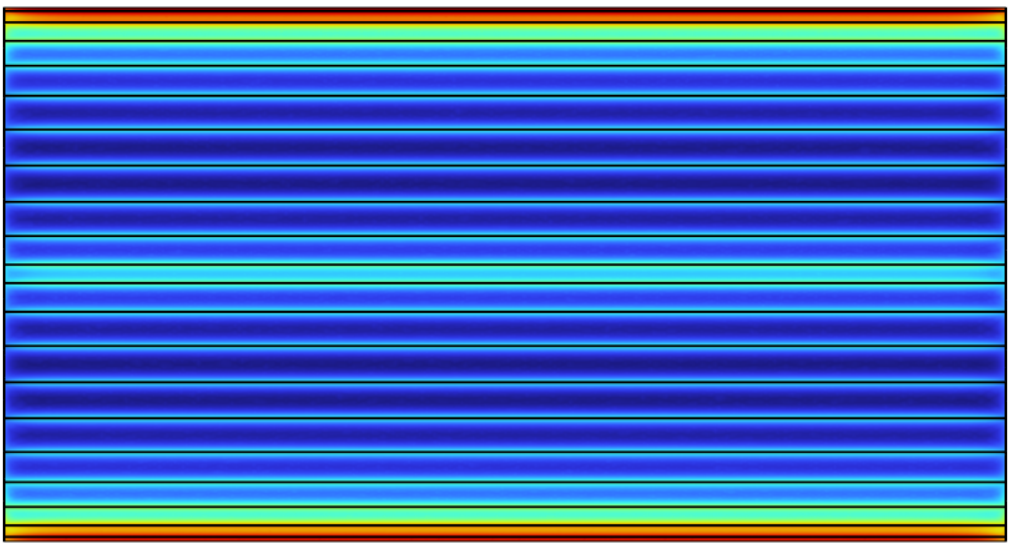


Figure 4.2: Steady-state temperature distribution for the recommended design: Pd–Ag alloy, 20 traces, $V_{\text{in}} = 12\,\mathrm{V}$. The minimum temperature ($288.2\,\mathrm{K}$) is well above the dew point ($283.15\,\mathrm{K}$).

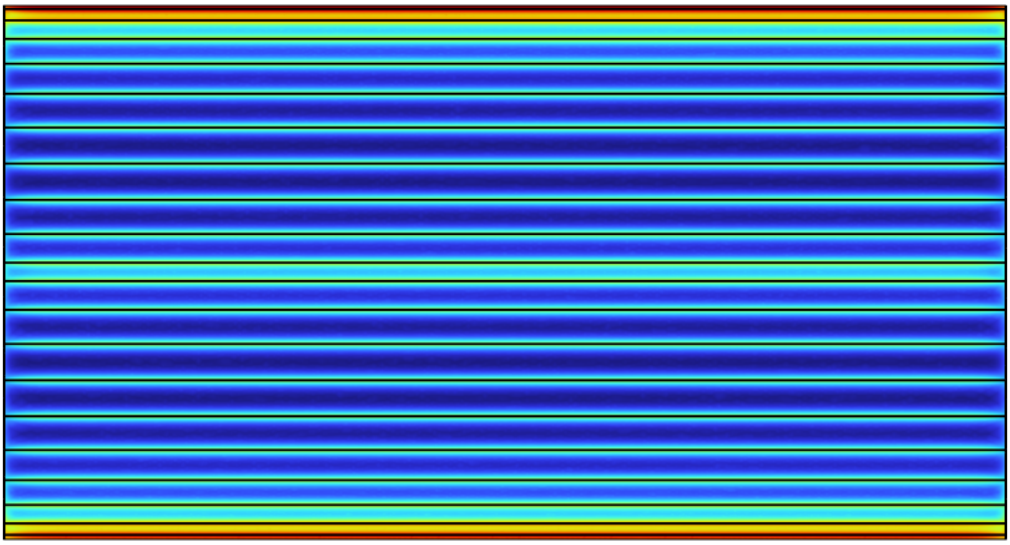


Figure 4.3: Temperature distribution at $t = 12\,\mathrm{min}$ (moment of defrost) for Pd–Ag alloy, 20 traces, $V_{\text{in}} = 12\,\mathrm{V}$.

4.3 Effect of Trace Material Conductivity

SEBTAIN?
A key finding of this study is the strong dependence of defroster performance on the trace electrical conductivity. The Joule heating power per unit length of trace scales as:

$$P \propto \frac{V_{\text{in}}^2 \cdot \sigma_{\text{eff}}}{L_{\text{trace}}} \tag{4.1}$$

where σ_{eff} is the effective 2D conductivity from Equation (1.1). This explains why:

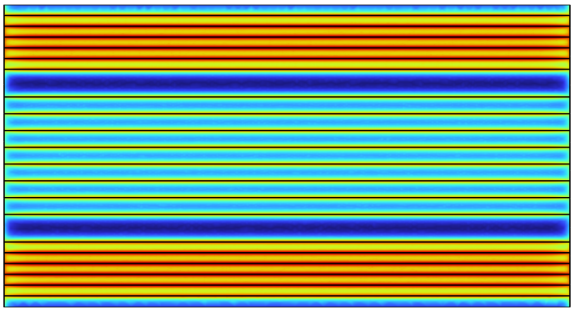
- Materials with very high conductivity (Silver: 5.75×10^7, Gold: 4.2×10^7 S/m) generate excessive heating, reaching temperatures of 500–600 K that would crack the glass.
- Materials with very low conductivity (Carbon ink: 10^4 S/m) generate negligible heating and cannot defrost the window.
- The optimal conductivity range for this geometry and voltage is $\sim 10^5$–10^6 S/m, which corresponds to thick-film paste materials (Pd–Ag alloy, Nichrome V).

TODO (Sebtain): Plot — T_{min} vs σ_{trace} (log scale), $n_{\text{traces}} = 20$, $12\,\mathrm{V}$

Figure 4.4: Effect of trace electrical conductivity on steady-state T_{min} ($n_{\text{traces}} = 20$, $V_{\text{in}} = 12\,\mathrm{V}$). The optimal range is highlighted, with Carbon ink too cold and bulk Silver/Gold overheating.

4.4 Trace Pattern Optimisation

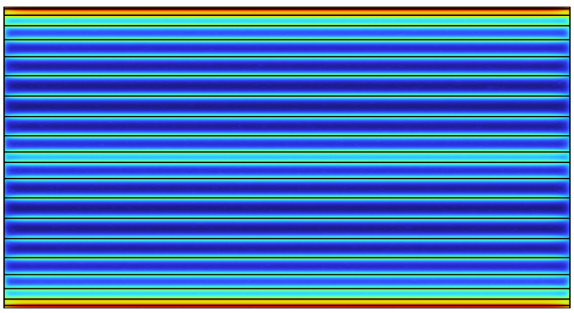
Two approaches to non-uniform trace spacing were analyzed: a discrete zoned layout (Approach A, Sim 4) and a continuous Chebyshev distribution (Approach B, Sim 5). Both use 20 traces of Pd–Ag alloy at $V_{\text{in}} = 12\,\text{V}$ to make sure the simulations are directly comparable to the uniform baseline (Sim 3).
The zoned configuration places 6 traces in the top $200\,\text{mm}$, 8 traces in the middle $400\,\text{mm}$ and 6 traces in the bottom $200\,\text{mm}$. This reduces the edge gap from $38.1\,\text{mm}$ (uniform) to $28.6\,\text{mm}$ while widening the centre gap to $44.4\,\text{mm}$.
The time-dependent simulation shows that this configuration defrosts in $11\,\text{min}$ (vs. $12\,\text{min}$ for uniform) and reaches a steady-state T_{min} of $289.3\,\text{K}$ (vs. $288.2\,\text{K}$). The improvement is modest: $1\,\text{min}$ faster defrost and $1.1\,\text{K}$ higher T_{min}.



TODO (Sebtain): Sim3 uniform 20 traces at defrost moment ($t = 12\,\text{min}$)

Figure 4.5: Temperature distribution at the moment of defrost. (a) Zoned 6/8/6 at $t = 11\,\text{min}$. (b) Uniform baseline at $t = 12\,\text{min}$. Both: Pd–Ag alloy, 20 traces, $12\,\text{V}$.

The Chebyshev approach (Equation (2.1)) is far more effective. With the optimal exponent $p = 1.3$ there is a better distribution of the traces on the rear window and so optimizing the heating.
THIS PICTURE IS ALREADY IN THE REPORT



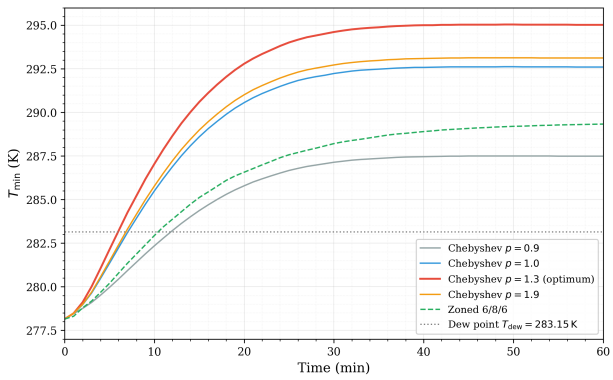


Figure 4.6: Chebyshev spacing results. (a) Temperature distribution at $t = 6\,\text{min}$ for $p = 1.3$. (b) Transient T_{min} evolution for selected p values; the dashed line is the dew point.

Table 4.3 summarises the performance of all trace patterns.

Table 4.3: Comparison of trace spacing approaches (Pd–Ag alloy, 20 traces, $V_{\text{in}} = 12\,\text{V}$).

Approach	**Defrost time**	**Steady-state T_{min} (K)**	**Improvement**
Uniform (baseline)	12 min	288.2	—
Zoned A2 (6/8/6)	11 min	289.3	-8% time, $+1.1\,\text{K}$
Chebyshev $p = 1.0$	8 min	292.6	-33% time, $+4.4\,\text{K}$
Chebyshev $p = 1.3$	**6 min**	**295.0**	**-50% time**, $+6.8\,\text{K}$

The Chebyshev $p = 1.3$ spacing halves the defrost time compared to uniform spacing while raising T_{min} by $6.8\,\text{K}$. This is achieved without any change in material, voltage, or number of traces — purely through optimised trace placement. The result demonstrates that **geometry optimisation** using COMSOL's parametric capabilities can be as impactful as material or electrical optimisation.

4.5 Discussion

4.5.1 Strengths

With the use of Chebyshev spacin the final design is able to achie a steady-state T_{min} of $295.0\,\text{K}$, $12\,\text{K}$ above the dew point and only in $6\,\text{min}$ compared to $12\,\text{min}$ with uniform spacing. This improvement comes only from trace placement, at zero additional cost. The thermal stresses remain within textsafe limits (safety factor $\sim 3.1\times$) and the design operates at nominal CHECK IF TRUE $12\,\text{V}$ battery voltage, meaning it works even with the engine off.

The parametric approach — sweeping material, trace count, voltage, and spacing exponent — provides a comprehensive design map that allows easy trade-off analysis for different requirements. The multiphysics coupling between Electric Currents, Heat Transfer, and Solid Mechanics captures the essential physics of the defroster operation, and the mesh convergence study confirms that results are mesh-independent for the recommended configurations. The boundary conditions (h_{ext}, h_{int}) are supported by published experimental data [5], and the sequential solve strategy for the stress analysis overcomes the COMSOL 6.2 spatial frame limitation while still capturing the thermal stress state. I DONT THINK THIS IS NECESSARY

4.5.2 Limitations, Future Work, and Broader Implications

In order to complete the analyses approximations have been made that lower the accuracy of the present analysis. The use of 2D planar geometry is not accurate as it fails to consider the actual curved shape of the rear windshield, which influences the heat flux and stress distribution; a 3D analysis would have been more realistic. The application of fixed boundary conditions at the glass edges is an overestimation of the maximum stress levels as realistically they would be lower. Using elastic springs instead of rigid supports to model the rubber gasket would produce more realistic results, since a rear window is supported by an elastic seal and the car frame. Heat transfer coefficients are kept fixed throughout the analysis, however, in reality, h_{ext} varies with vehicle speed due to convection, while h_{int} depends on the cabin airflow conditions set by the internal air conditioning system. Using variable climatic conditions and varying T_{amb}, T_{dew}, and h_{ext} would give more realistic results. Furthermore, the material properties used were referenced from the EduPack database for bulk materials. The thick-film paste used in screen printing may have different properties due to the manufacturing process and microstructure, which could also affect the heating performance.

The Chebyshev spacing result has broader implications beyond the rear window heating trace pattern. The key insight is that non-uniform spatial distribution of heat sources, optimised using a single continuous parameter, can be as effective as changing materials or operating conditions. This principle applies to any distributed heating or cooling problem where temperature uniformity has to be achieved.

STILL DO: ELEMENT ORDER CONVERGENCE ANALYSIS — comparing linear vs. quadratic elements for the Joule heating/temperature solution to verify that the default element order is sufficient.

STILL DO: ADAPTIVE MESH REFINEMENT ANALYSIS — using COMSOL's adaptive mesh refinement to verify error control near the trace boundaries where temperature gradients are steepest.

Chapter 5

Conclusion

This report presents the design and optimisation of a resistive rear window defroster using coupled multiphysics finite element modelling in COMSOL Multiphysics 6.2. The design was developed iteratively through five simulations addressing thermal performance, material selection, structural integrity, transient response, and trace pattern optimisation.
The key findings of this study are:

1. **The number of heating traces is the most influential design parameter.** Increasing the trace count from 12 to 20 raises T_{min} by over $6\,\mathrm{K}$ at $12\,\mathrm{V}$, whereas increasing voltage from 12 to $14.5\,\mathrm{V}$ at 14 traces yields only $1.2\,\mathrm{K}$.

2. **Pd–Ag alloy ($\sigma = 2.3 \times 10^6\,\mathrm{S\,m^{-1}}$) is the optimal trace material.** It reaches the dew point with 14 traces at $12\,\mathrm{V}$, while Nichrome V requires 18–20 traces. Bulk conductivity materials (Silver, Gold) overheat catastrophically and are not viable for this geometry.

3. **Both shortlisted materials pass the stress analysis** with safety factors exceeding $3\times$, confirming that thermal expansion mismatch between the traces and glass does not pose a structural risk.

4. **The recommended design** is:
   - Material: Pd–Ag alloy
   - Number of traces: 20
   - Voltage: $12\,\mathrm{V}$ (nominal battery)
   - Steady-state T_{min}: $288.2\,\mathrm{K}$ ($5\,\mathrm{K}$ above dew point)
   - Defrost time: 12 minutes
   - Stress safety factor: $\sim 3.1\times$

5. **Alternative high-performance design:** Pd–Ag alloy, 22 traces, $14.5\,\mathrm{V}$ — achieves 8-minute defrost time when the engine is running.

6. **Chebyshev trace spacing halves the defrost time.** The non-uniform Chebyshev spacing (Equation (2.1)) with exponent $p = 1.3$ reduces the defrost time from $12\,\mathrm{min}$ to $6\,\mathrm{min}$ and raises T_{min} by $6.8\,\mathrm{K}$ — without changing the number of traces, material, or voltage. The discrete zoned approach (6/8/6) provides only a modest $1\,\mathrm{min}$ improvement, confirming that a continuous optimisation is needed. The optimal exponent ($p = 1.3$) was found via a 16-point parametric sweep, exploiting COMSOL's ability to parametrically update the geometry.

7. **Final recommended design:**
   - Material: Pd–Ag alloy ($\sigma = 2.3 \times 10^6\,\mathrm{S\,m^{-1}}$)
   - Number of traces: 20
   - Spacing: Chebyshev with $p = 1.3$
   - Voltage: $12\,\mathrm{V}$ (nominal battery)
   - Steady-state T_{min}: $295.0\,\mathrm{K}$ ($12\,\mathrm{K}$ above dew point)
   - Defrost time: $6\,\mathrm{min}$
   - Stress safety factor: $\sim 3.1\times$

- Window obscuration: 4.25% ($< 5\%$ SAE J216 limit)

The simulation methodology — coupling Electric Currents, Heat Transfer, and Solid Mechanics with parametric sweeps over material properties, geometry, and operating conditions — provides a systematic framework for defroster design. The model was validated through mesh convergence, back-of-envelope analytical estimates (agreeing to within 0.03%), and comparison with published experimental data.

The key insight of this study is that **geometry optimisation** (trace placement) can be equally or more impactful than material or electrical optimisation. The Chebyshev spacing approach provides a mathematically principled, single-parameter design specification that can be applied to any rectangular heated-glass application. Future work should extend the model to 3D curved glass, validate against physical prototypes, and perform weather condition sweeps to test design robustness across climate scenarios.

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